

# Peter Afif

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Senior Computer Science student (Expected Aug 2026) seeking full-stack and ML/AI roles; open to remote/relocation.

Shipped 3 deployed apps with Python, Scikit-learn, React, Node.js, PostgreSQL, pgvector, and LLM/RAG pipelines.

## EDUCATION

### California State University, Fullerton

*Bachelor of Science in Computer Science, GPA: 3.2*

Fullerton, CA

*Expected July 2026*

### Orange Coast College

*Associate's in Computer Science, GPA: 3.2*

Costa Mesa, CA

*Aug. 2021 – May 2024*

## PROJECTS

### ResumeShift ↗ | Full-Stack | React, Node.js, Express, GPT-4o, OpenAI API | Deployed

- Engineered a 5-phase stateful AI resume analyzer using **GPT-4o** across a **multi-turn conversation** (goal extraction, JD grounding, line-level analysis, bullet rewrites, and adjacent role matching) with a **React/Vite** frontend and **Node.js/Express** backend
- Designed a custom **eval framework** measuring **keyword hit rate** across 4 **prompt engineering** iterations; improved score from **0.10 to 0.81** through targeted prompt changes and eval methodology fixes across 10 labeled test cases
- Integrated a **GPT-4o-mini** intent classifier to replace brittle regex, enabling automatic **phase chaining** server-side; implemented optional **JD grounding** to extract role-specific competency phrases absent from the resume

### Groove Vault ↗ | Full-Stack + RAG | React, TypeScript, Node.js, PostgreSQL, OpenAI API, GitHub Actions

- Built a **RAG**-powered drum technique assistant: PDFs chunked, embedded with **OpenAI text-embedding-3-small**, stored in **pgvector** (PostgreSQL), and retrieved using **cosine similarity**, with **GPT-4o** generating grounded responses with source attribution; rate-limited chat API (20 req/15 min)
- Built a **RESTful API** on **PostgreSQL** using a **window function CTE** (`ROW_NUMBER() OVER PARTITION BY`) for per-rudiment BPM trend deltas and **gap-and-island SQL** for accurate practice streak calculation
- Engineered a full-stack **TypeScript** app (**React**, **Express.js**) with **JWT** + **Google OAuth** auth and **Web Audio API** metronome; configured **CI/CD** with 5 parallel **GitHub Actions** jobs (**Vitest** + **Supertest** against a fully mocked database)

### Backbeat AI ↗ | Full-Stack + ML | Python, Scikit-learn, Librosa, React, Node.js, PostgreSQL, Docker

- Built a **Python** audio classification pipeline using **Librosa** to extract 16 rhythm features (IOI statistics, beat deviation, tempo delta) and a **Scikit-learn GradientBoostingClassifier** to classify live drumming into 4 timing states
- Resolved heavy class imbalance by synthesizing 57k augmented training samples and applying **class-weighted training**, improving minority class recall from 0.16 to 0.45
- Deployed a **React**, **Express.js**, and **PostgreSQL** application via **Docker**, engineering server-side **FFmpeg** audio conversion and **WebSocket** connections for real-time prediction streaming

## EXPERIENCE

### Music Instructor

*McCormick Music, Kidz Rock, Drum Path*

2019 – Present

*Orange County, CA*

- Designed and delivered structured curricula for **50+ students** across private and group settings, tracking individual progress and adapting lesson plans based on measurable performance data

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, SQL, HTML, CSS

**Frameworks:** React, Node.js, Express.js, Tailwind CSS, Vite

**Databases:** PostgreSQL, pgvector

**Developer Tools:** Git, GitHub Actions, Docker, Linux

**Libraries:** OpenAI API, Scikit-learn, Librosa, JWT, REST APIs, Vitest, Supertest

**Concepts:** RAG, LLMs, Prompt Engineering, Vector Search, Machine Learning, Feature Engineering, CI/CD, Data Structures & Algorithms, OOP

**Languages:** French, English, Arabic